NABIEL MUHAMMAD IMJAUZANANSYAH

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PROFESSIONAL SUMMARY

Fresh Information Systems graduate (GPA 3.66/4.00) specializing in applied machine learning and AI systems. Built and deployed a CNN-LSTM computer vision model achieving 96% validation accuracy for real-time sign language translation, published in Jurnal ICT. Hands-on experience with LSTM time-series forecasting, LLM/agentic integration (Google Gemini API), and NLP classification through independent projects and the DBS Foundation Machine Learning Engineering program. Comfortable working end-to-end: data collection, model training and evaluation, and production deployment behind REST APIs.

EDUCATION

Bachelor of Information Systems, Universitas Muhammadiyah Sumatera Utara (UMSU)

Medan, Indonesia | 2022 - June 2026

- GPA: 3.66 / 4.00
- Final year project: real-time BISINDO sign language translator using CNN-LSTM and Text-to-Speech, achieving 96% validation accuracy
- published in Jurnal ICT: Information and Communication Technologies (2026)
- Relevant coursework: Machine Learning, Data Analysis, Database Systems, Algorithm and Programming, Software Engineering, Web Development

RELEVANT EXPERIENCE

Machine Learning Engineer Trainee, DBS Foundation Coding Camp 2025

Remote, Part-time | Feb 2025 - Aug 2025

- Completed an intensive machine learning engineering program covering data preprocessing, model development, evaluation, and deployment workflows
- Built and evaluated data-driven prototypes, including a recommendation system and an NLP-based text classification model, using Python, Scikit-learn, TensorFlow, and Keras
- Practiced reproducible project structuring, technical documentation, and deployment preparation to support production-ready ML workflows

Software Developer Intern, PT Telekomunikasi Selular (Telkomsel)

Medan, Indonesia | Jun 2025 - Sep 2025

- Designed and built a Python-based Telegram voucher-distribution bot (Bot Voucher BGA SUMBAGUT) integrated with the internal SAMOSIR.LIVE platform via REST APIs and real-time webhooks, automating reward redemption, challenge, survey, and point management for the BGA Regional SUMBAGUT network
- Engineered end-to-end encryption, two-factor authentication, and fraud-detection safeguards, and architected the system to scale from 100 to 1,000 transactions per hour during peak periods
- Deployed the bot to production in September 2025, achieving 100% completion of the assigned scope (exceeding the initial 80% target) and a 98/100 supervisor evaluation score; the bot has remained in continuous operation for 10+ months since, well beyond the internship period
- Enhanced modules within a Laravel-based internal web application, including reward management and administrative functionality, while managing data interactions with MySQL

PROJECTS

NzSignify - Real-Time BISINDO Sign Language Translator

Final Year Project | Dec 2025 - Apr 2026

- Developed a real-time Indonesian Sign Language (BISINDO) translation system using a hybrid CNN-LSTM deep learning architecture for gesture classification
- Collected and preprocessed an original dataset of 11,000 image frames across 11 gesture classes from 3 participants, applying spatio-temporal normalization and 30-frame sequence extraction
- Trained and optimized the model in Google Colab, achieving 96% classification accuracy, with confidence-threshold filtering to improve prediction reliability

- Built a React.js frontend integrated with a Node.js backend to process real-time webcam streams and communicate with the AI inference service via REST API
- Implemented Text-to-Speech (TTS) functionality to convert recognized gestures into spoken output, improving accessibility for end users

InvestMentor - AI-Powered Trading Simulator & Agentic Mentor Platform

Full-Stack AI Engineer | Independent Project | Sep 2025 - Jun 2026

- Engineered an interactive AI investment mentor using the Google Gemini 2.0 Flash SDK, implementing multi-turn conversational memory and context-aware responses for stock market analysis
- Deployed an LSTM-based machine learning model using TensorFlow.js to generate daily stock price predictions with calculated confidence levels
- Built real-time data ingestion pipelines integrating the Yahoo Finance API and Google News RSS, serving dynamic market context to the frontend via WebSocket (Socket.IO)
- Developed a Node.js/Express.js backend with a secure REST API architecture, implementing API rate limiting, JWT authentication, and database transactions via Sequelize ORM

Additional projects: ParentCare - AI-powered parenting style platform (Random Forest classifier, team of 6, DBS Foundation Capstone, 2025); Fashion Studio ETL Pipeline - independent Python data pipeline (Mar 2025).

TECHNICAL SKILLS

AI & Machine Learning: TensorFlow, TensorFlow.js, Keras, Scikit-learn, CNN-LSTM, LSTM, NLP, Computer Vision, LLM/Generative AI Integration (Google Gemini API), Model Evaluation & Deployment

Programming Languages: Python, JavaScript, Node.js, PHP

Data Engineering & Analytics: ETL Pipelines, Real-Time Data Pipelines, Pandas, NumPy, Data Cleaning & Validation, REST API Integration, WebSocket (Socket.IO), Google Sheets API

Backend & Database: Express.js, Laravel, JWT Authentication, Sequelize ORM, MySQL, PostgreSQL, SQL Fundamentals, Relational Database Design

Tools & Platforms: Git, GitHub, Docker, Google Colab, VS Code, Vercel, Railway

CERTIFICATIONS

- DBS Foundation Coding Camp: Machine Learning Engineering - Dicoding Indonesia (2025)
- Applied Machine Learning, Machine Learning Development, Machine Learning for Beginners - Dicoding & Google Developers (2025)
- Data Analysis with Python - Dicoding & Google Developers (2025)
- Telkomsel Official Internship Certificate - Telkomsel BGA Regional SUMBAGUT (2025)
- Additional: SQL Fundamentals, Git and GitHub Fundamentals, Fundamentals of Data Processing/ETL Pipelines - Dicoding & Google Developers (2025)